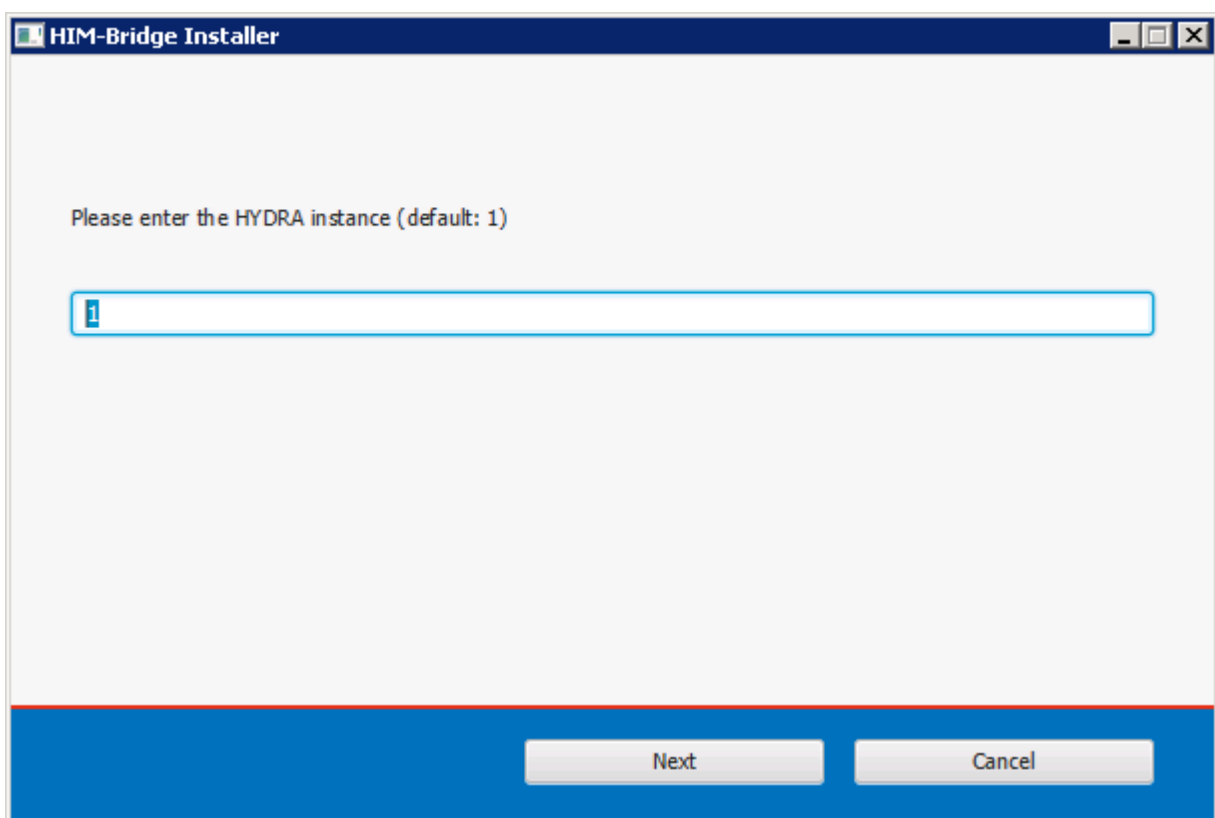


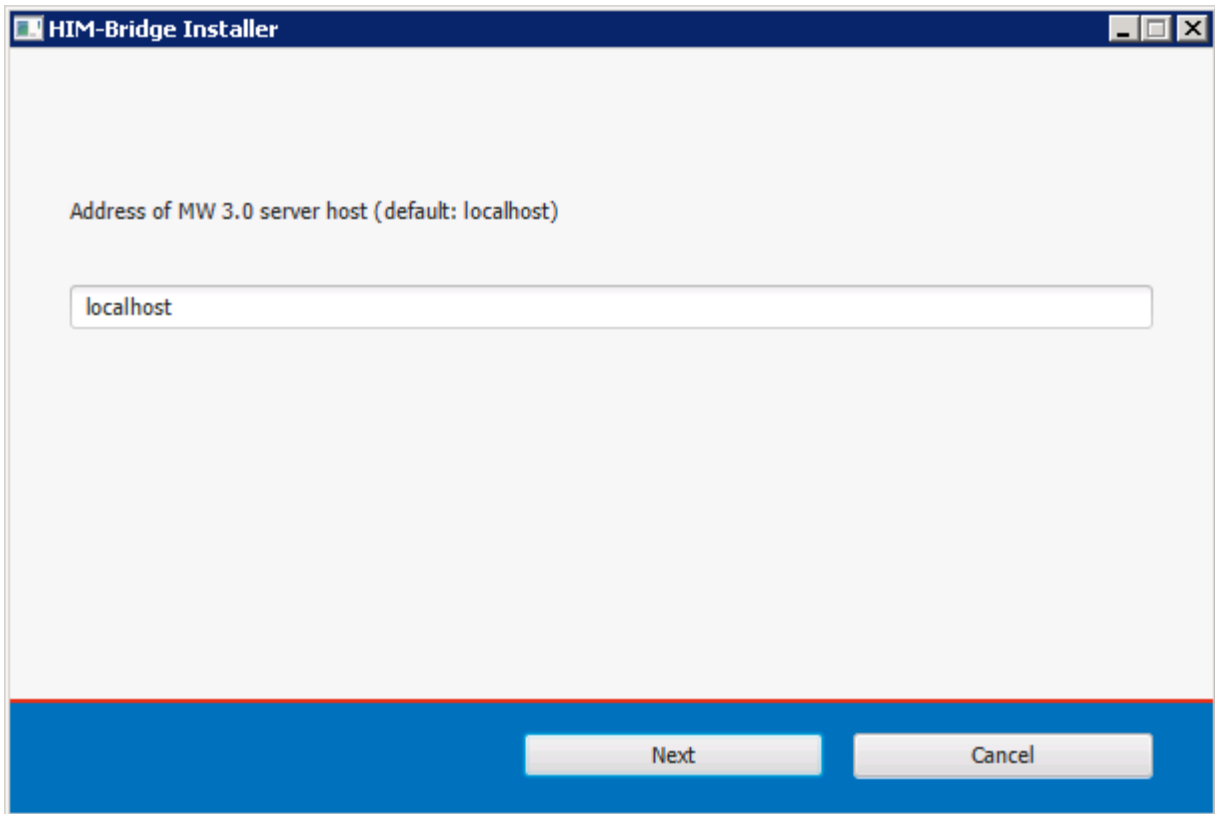
1 WRM-TDA Configuration

1.1 Installation HIM-Bridge

- You must install the HIM-Bridge in the HYDRA server.
- Unpack the zip-file anywhere on the HYDRA server.
- Start the setup as the administrator using the file installer.exe.
- Enter the HYDRA instance number (in case of single-instance systems always 1 / with other systems enter the number of the instance the bridge is installed for).



The host of the HYDRA server is localhost as the installation is run in the same system.



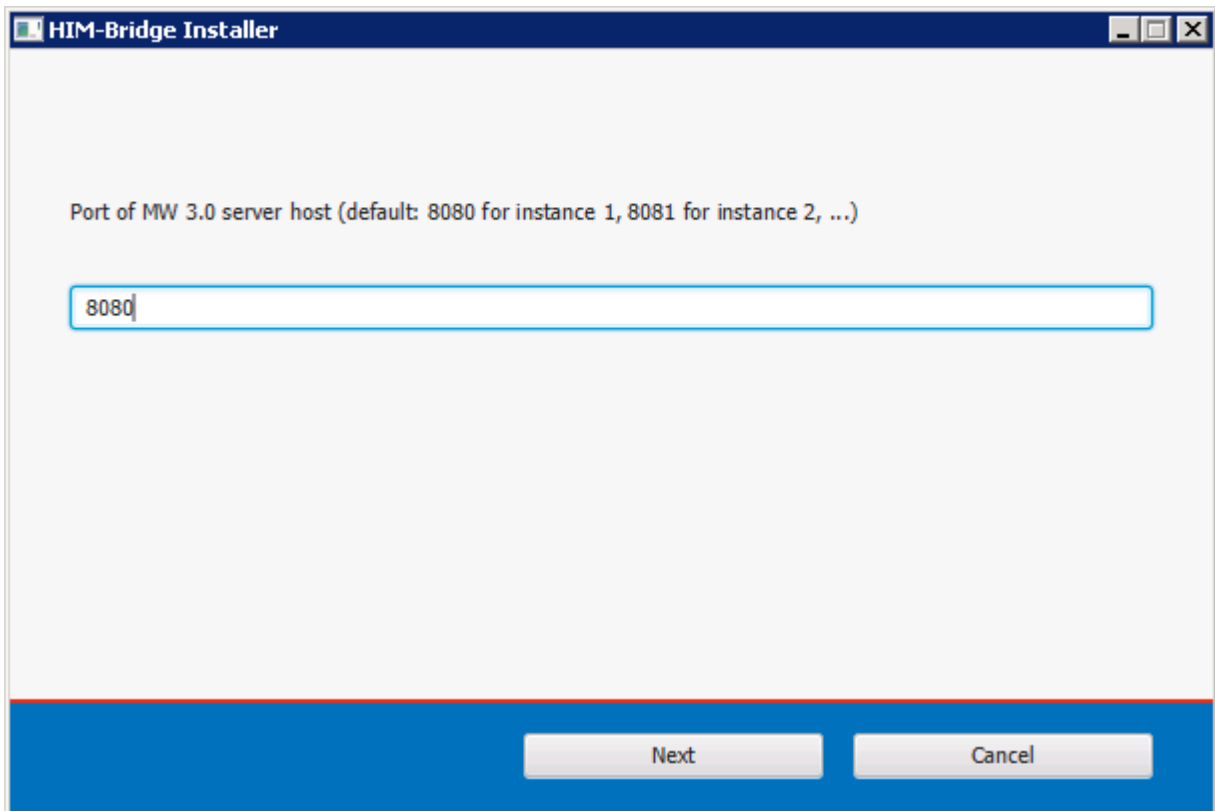
HIM-Bridge Installer

Address of MW 3.0 server host (default: localhost)

localhost

Next Cancel

Enter the Tomcat port of the instance (usually 8080 for instance 1, 8081 for instance 2,... If you do not know the port, you can look it up in the system list of the MOC logon window).



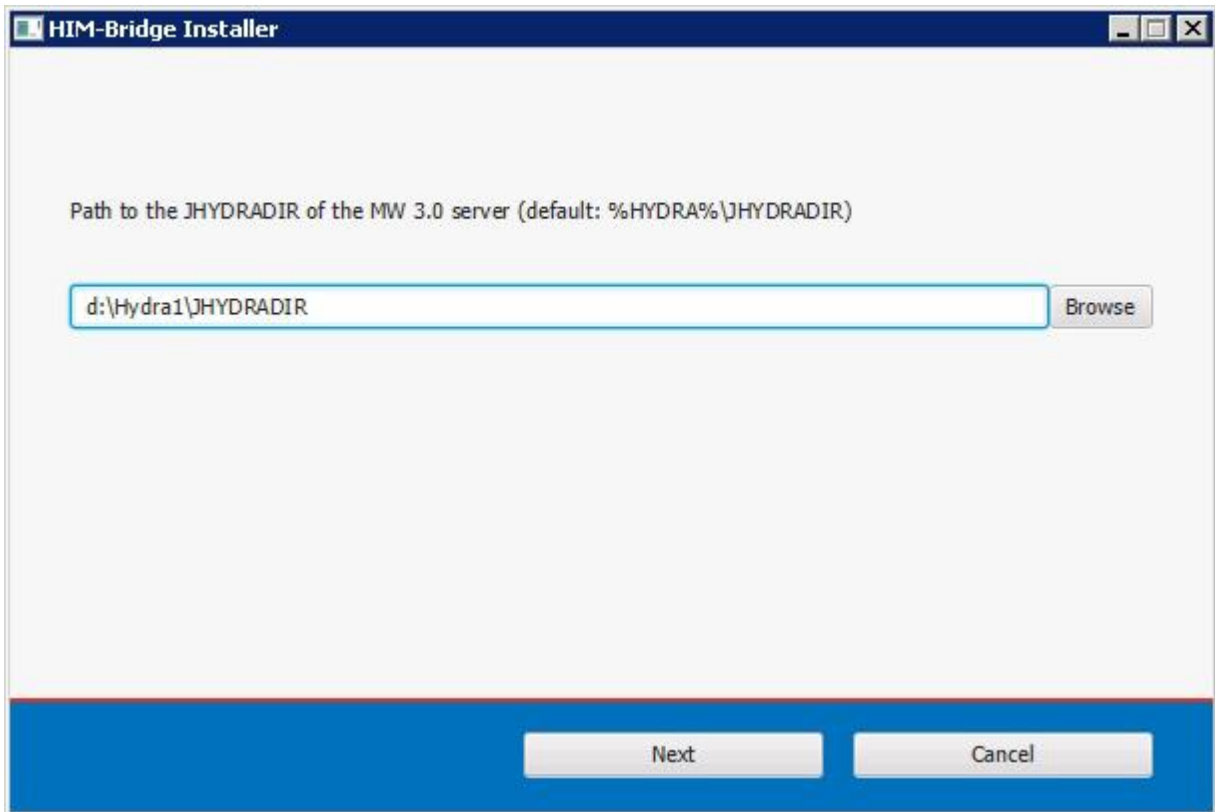
HIM-Bridge Installer

Port of MW 3.0 server host (default: 8080 for instance 1, 8081 for instance 2, ...)

8080

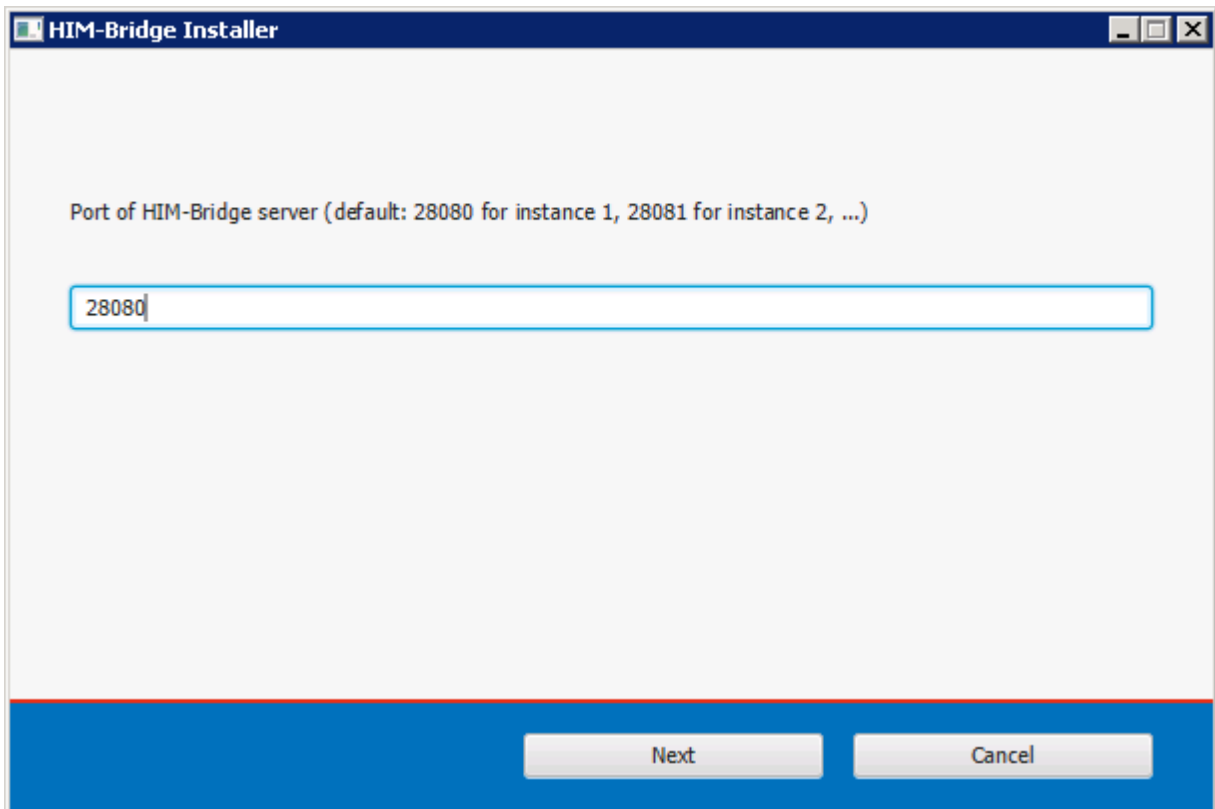
Next Cancel

Enter the path to JHYDRADIR (usually the folder JHYDRADIR or JHYDRADIRX in the HYDRA directory)



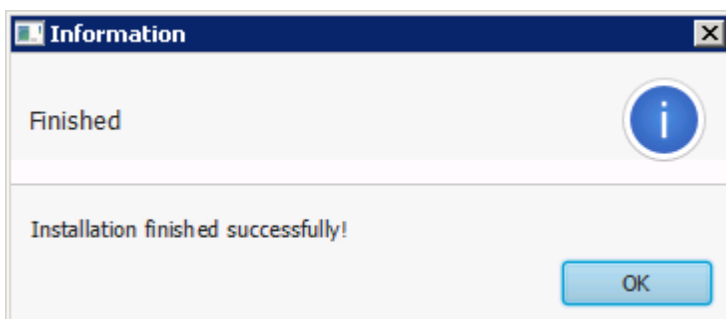
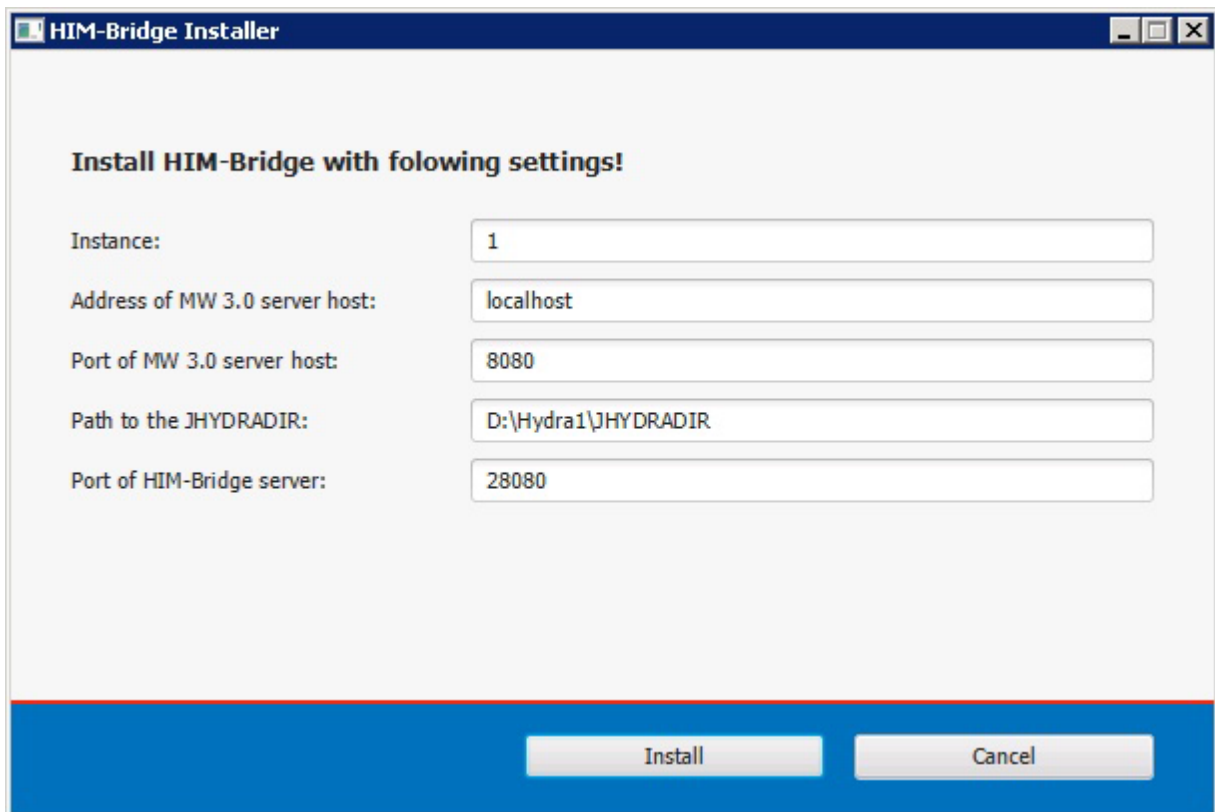
The screenshot shows the 'HIM-Bridge Installer' window. The title bar is dark blue with the text 'HIM-Bridge Installer' and standard window controls. The main area is light gray. It contains the text 'Path to the JHYDRADIR of the MW 3.0 server (default: %HYDRA%\JHYDRADIR)'. Below this is a text input field containing 'd:\Hydra1\JHYDRADIR' and a 'Browse' button to its right. At the bottom, there is a blue bar with two buttons: 'Next' and 'Cancel'.

Enter the port of the HIM-Bridge (usually the Tomcat port + 20000).

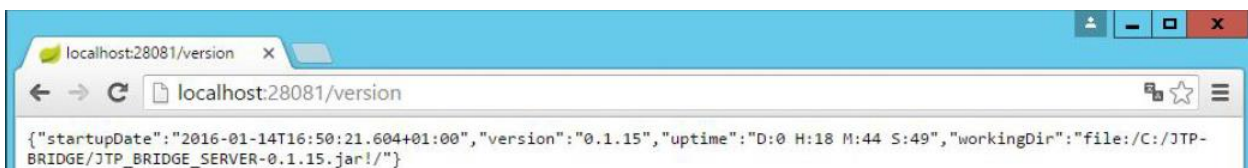


The screenshot shows the 'HIM-Bridge Installer' window. The title bar is dark blue with the text 'HIM-Bridge Installer' and standard window controls. The main area is light gray. It contains the text 'Port of HIM-Bridge server (default: 28080 for instance 1, 28081 for instance 2, ...)'. Below this is a text input field containing '28080'. At the bottom, there is a blue bar with two buttons: 'Next' and 'Cancel'.

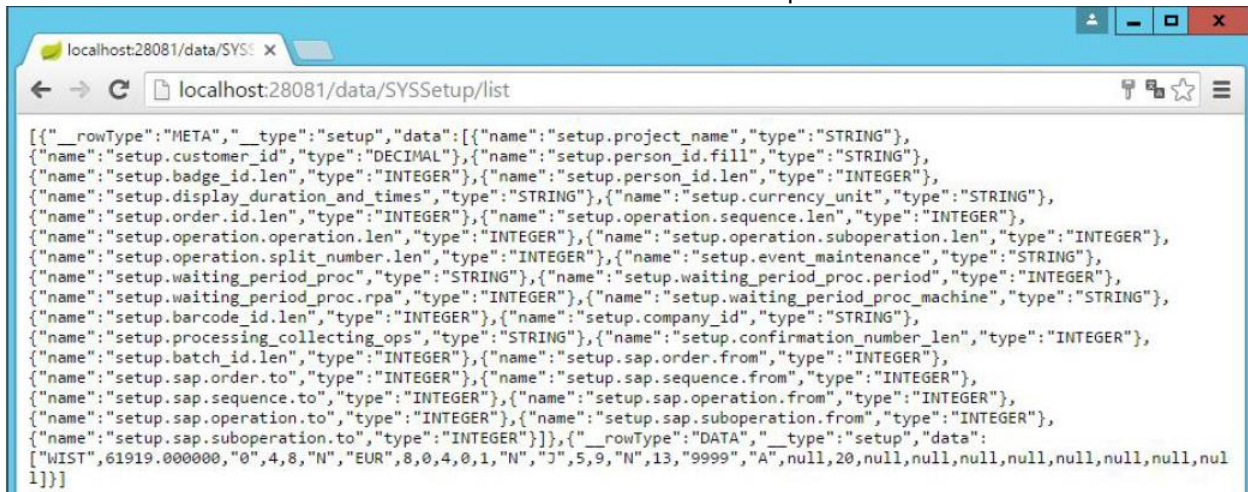
Confirm the data overview and finish the installation.



Test the HIM-Bridge by calling the following URL in a browser in the HYDRA server: <http://localhost:<port of HIM bridge>/version> (e.g. <http://localhost:28081/version>).



Test the connection between the HIM-Bridge and HYDRA by calling the following URL in a browser in the HYDRA server: `http://localhost:<port of HIM bridge>/data/SYSSetup/list`
Please note: Use a valid MOC user when asked to enter user and password.



1.2 Installation MDC (Machine data collector)

Unpack the zip file to the install location selected for MDC.

Open a prompt as administrator.

Go to the MDC installation directory. Install MDC as Windows service by using the following command (replace HYDRA-X with the instance number - e.g. HYDRA-1-MDC):

```
bin\mpdv.MachineDataCollector.Console.exe --install HYDRA-X-MDC
```

Copy the file

```
<MDC directory>\bin\config\config_balluff.xml
```

to

```
%ProgramData%\mpdv\mdc\config_HYDRA-X-MDC.xml
```

(replace HYDRA-X-MDC with the name used above)

Open the configuration file previously created in an editor.

Set the port on which the http server listens to requests from Mold-ID. You can find the port in row 6 of the configuration. The default port is 30080. In case of several HYDRA instances, use e.g. 30080 for instance 1, 30081 for instance 2, etc.

```
<Parameter Name="Config_Port" Value="30080" />
```

Configure the host / port that includes the HIM-Bridge of the HYDRA instance in row 9 of the configuration (host name of the HYDRA server and port as assigned in chapter 1.1 on installing the HIM-Bridge).

```
<Parameter Name="RemoteEndpoint" Value="http://scc7:28081" />
```

Configure user name and password of an MOC user used for calling HYDRA web services. The user must be available in HYDRA.

Please note: The encrypted password is stored in the configuration file. An MDC tool is used to encrypt the password. You can call the tool in the command line as follows:

```
mpdv.MachineDataCollector.Console.exe --encode Password
```

```
<Parameter Name="UserName" Value="12345" />
```

```
<Parameter Name="Password" Value="J1FuIQ==" />
```

Start the created service.

1.3 Required HYDRA master data

Make sure that all tags connected via Mold-ID exist in HYDRA as a resource. Only then, the data can be transferred into the system.

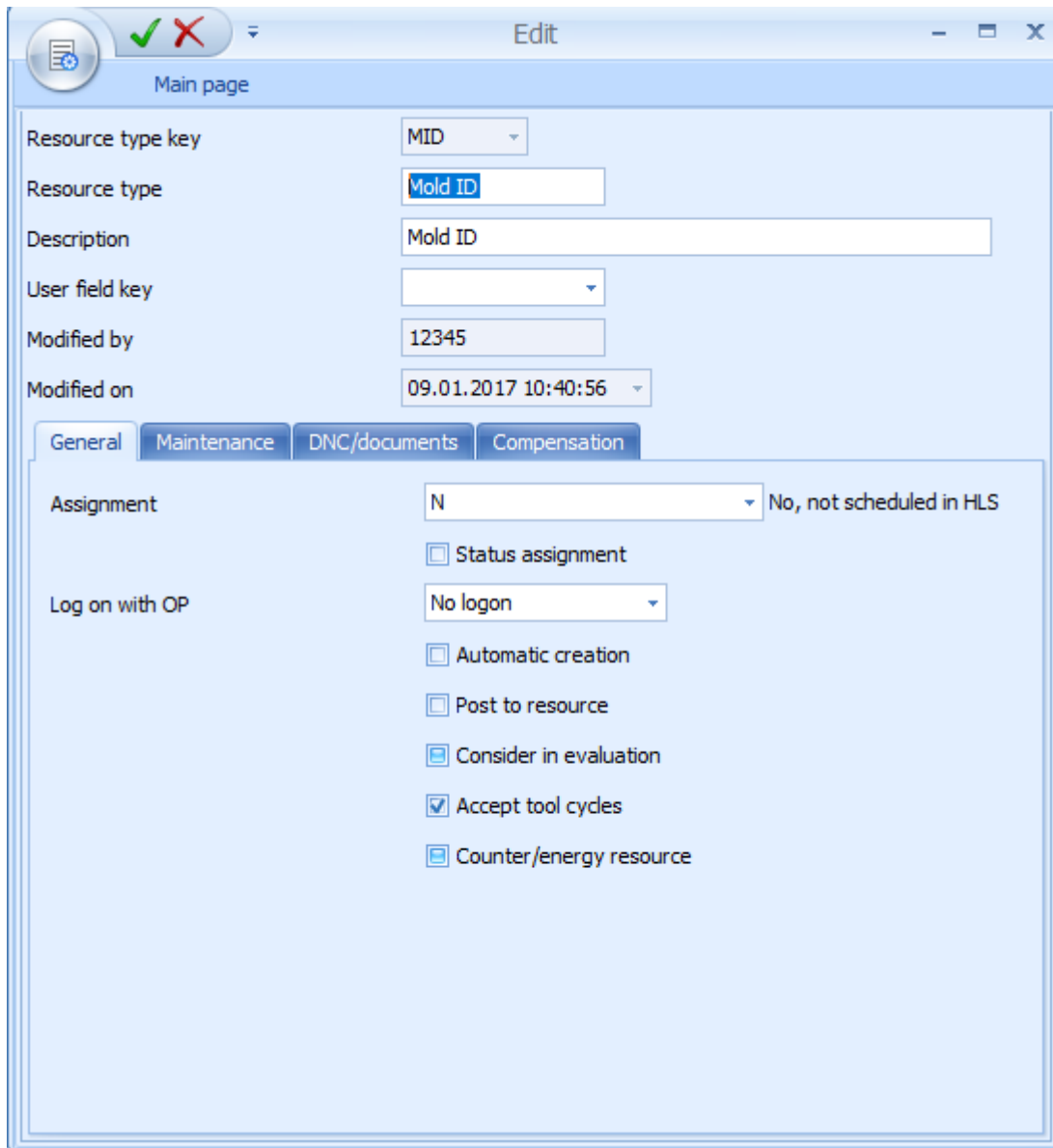
The tags attached via Mold-ID only have a "resource number" (form name). To identify a resource, HYDRA actually uses a resource type + resource number.

For this reason, the resource type used with Mold-ID is permanently defined in the system.

1.3.1 Defining a resource type that will be used for all Mold-ID tags

You can use e.g. the type MID as resource type for Mold-ID tags. Make the settings as displayed in the screenshot.

Master data -> Resources -> Resource types



Resource type key: MID

Resource type: Mold ID

Description: Mold ID

User field key:

Modified by: 12345

Modified on: 09.01.2017 10:40:56

General Maintenance DNC/documents Compensation

Assignment: N No, not scheduled in HLS

Log on with OP: No logon

☐ Status assignment

☐ Automatic creation

☐ Post to resource

☒ Consider in evaluation

☒ Accept tool cycles

☒ Counter/energy resource

1.3.2 Defining statuses for the resource type

A tag can have 4 statuses. All 4 statuses must exist in HYDRA for the resource type that will be used. One status must be configured as release status.

Define the 4 statuses as displayed in the screenshots.

Master data -> Resources -> Resource status

Resource type key	MID
Resource family	...
Status	97
Designation	Instandhaltung
Color	
Authorization	0
Material buffer	...
Processing	
Assignment	No assignment
Collection	Released
☐ Show resource with this status on terminal	

Resource type key	MID
Resource family	...
Status	98
Designation	Reparatur
Color	
Authorization	0
Material buffer	...
Processing	
Assignment	No assignment
Collection	Released
☐ Show resource with this status on terminal	

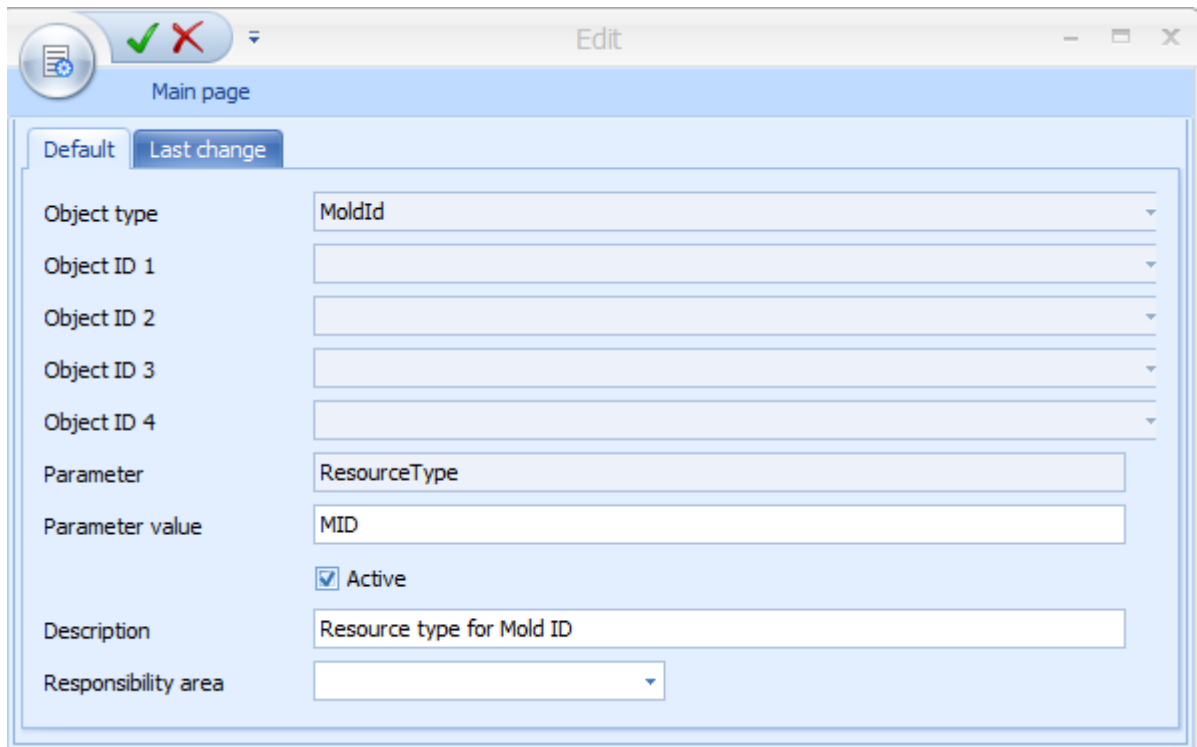
Resource type key	MID
Resource family	...
Status	99
Designation	Gesperrt
Color	
Authorization	0
Material buffer	...
Processing	
Assignment	No assignment
Collection	Blocked
☐ Show resource with this status on terminal	

Resource type key	MID
Resource family	...
Status	999
Designation	FREIGEgeben
Color	Green
Authorization	0
Material buffer	...
Processing	Release
Assignment	No assignment
Collection	Released
☒ Show resource with this status on terminal	

1.3.3 Configuration of the resource type for Mold-ID tags

Define the resource type that will be used for Mold-ID tags via the *Advanced object configuration*.

System administration -> System settings -> Advanced object configuration



Edit

Main page

Default Last change

Object type MoldId

Object ID 1

Object ID 2

Object ID 3

Object ID 4

Parameter ResourceType

Parameter value MID

☒ Active

Description Resource type for Mold ID

Responsibility area

1.3.4 Creating the resource (per Mold-ID tag)

Create a resource with the previously defined type for each tag in HYDRA.

Master data -> Resources -> Resource configuration

The screenshot shows a web application window titled 'Edit' with a 'Main page' header. Below the header are four tabs: 'General', 'Resource configuration', 'User fields', and 'Commentary'. The 'Resource configuration' tab is active, displaying a form with various fields for resource management. The fields are organized into two main sections: a top section for general resource details and a bottom section for supplier information.

Field	Value
Resource type	MID
Resource	HYDRA01
Short name	
Designation	Mold ID tag HYDRA01
Responsibility area	--DEFAULT--
Cost center	
Inventory number	
Engraving number	
Drawing number	
Manufacturer	
Owner	
Acquisition date	
Acquisition costs	0,00 EUR
Storage loc.	
Delivery date	
Start-up date	
Guaranteed date	
External designation	
Resource type designation	
Usage	
Purch. order no.	

Supplier	
Supplier No.	
Supplier designation	
Supplier Address 1	

1.4 Restrictions

The *last device name* of the Mold-ID tag is stored in HYDRA as *Current storage location* of the resource. The value must not exceed 12 characters. If the value is longer, an error occurs and data cannot be transferred.

1.5 Configuration in Mold-ID

Mold-ID needs to be configured to send data to the MDC server via http. You can make the configurations in the web interface of Mold-ID.

Setup -> Mold ID

Check the option "Network Reporting: http".

- ➔ IP address is the IP address of the system running MDC.
- ➔ Port number is the port configured as "Config_Port" in MDC.
- ➔ Path is the URL. Enter permanently the value "/MoldId/SendRfidData/" here.

BALLUFF**MOLDID**

 Home

 Logout

 Setup ▾

 Info

Mold ID Configuration

Trigger debouncing (milliseconds)

Trigger Polarity

The number of milliseconds that a trigger must be active before it will be recognised as such. This is required to avoid false positives from line noise.

Which state is to be recognised as the start of a trigger.

Trigger state
Current trigger state: idle

Smartlight Brightness
0% (off) 100%
Sets how bright the Smartlight should be.

☒ Network Reporting: HTTP ▾

IP address

Port number

Path

☒ Save